

BORONG XU

Research Assistant at SLAI

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ABOUT ME

- Research Assistant at the Shenzhen Loop Area Institute (SLAI) and a top 3% graduate of HKUST's Computer Engineering (CPEG) program. Passionate about robotics and embodied intelligence, with research experience spanning visual-inertial odometry, SLAM, multi-camera localization, and quadruped control, as well as deep learning, vision transformers, and LLM-based agent systems. Eager to push the boundaries of autonomous systems in real-world deployments.

EDUCATION

The Hong Kong University of Science and Technology (HKUST)

Hong Kong SAR

BACHELOR OF ENGINEERING IN COMPUTER ENGINEERING

Graduated: June 2026

- GPA 3.84/4.3 (Top 3% among 157 students)
- Major GPA 4.03/4.3

High School Affiliated to Renmin University of China (RDFZ)

Beijing, China

HIGH SCHOOL DIPLOMA(GAOKAO)

Graduated June 2022

- Activities:
 - FIRST Robotics Club, Chief Engineer (season 2021)
 - FIRST Robotics Club, Alumni Mentor (season 2022-2024)

EXPERIENCE

Shenzhen Loop Area Institute (SLAI)

Shenzhen, China

RESEARCH ASSISTANT

Jun 2026 - Present

- Research on robotics and embodied intelligence.

PROJECT EXPERIENCE

LEGO Assembly Understanding

Supervised by Prof. Ziqi Wang

ONGOING RESEARCH PROJECT (HKUST × SLAI)

Mar 2026 - Present

- Building a pipeline that automatically generates step-by-step assembly trees for LEGO models: parsing visual assembly manuals into step graphs, then learning a repeat-aware perception policy over a LoRA-adapted DINOv2 backbone to decompose a target model into per-step part subsets, combined with search-based planning.

ELEC5660 Introduction to Aerial Robotics

Supervised by Prof. Shaojie Shen

COURSE PROJECT

Feb 2026 - May 2026

- Developed aerial robot algorithms from scratch, including visual-inertial odometry, Extended Kalman Filter (EKF) state estimation, and autonomous path planning within a simulation environment, enabling robust UAV localization, mapping, and navigation.

Quadruped Robot Control

Supervised by Prof. Maurice Pagnucco and Prof. Yang Song

RESEARCH PRACTICUM

Feb 2025 - Aug 2025

- Built and programmed a Unitree GO2 quadruped robot to perform complex tasks, integrating depth camera for accurate object localization (localization error < 10cm) and LiDAR for SLAM.

Underwater Localization System

Supervised by Prof. Huan Yin and Prof. Fumin Zhang

FINAL YEAR PROJECT

Jun 2025 - May 2026

- Developed a Multi-Camera Vision-Based Localization System for Tracking Multiple Autonomous Underwater Vehicles (AUVs). Reduced localization error in complex underwater lighting scenarios, improving positioning accuracy to within 10cm in a 1m×1m range.

HONORS

2022-2026 Dean's List, 5 times

HKUST

2023-2026 Continuing Undergraduate Students Scholarship, 10,000 HKD per year

HKUST

2023 Bronze Medal, The Innovation Competition: Our Livable City, We Engineer

HKIE

2022 Entry Scholarship, 70,000 HKD

HKUST



SKILLS

- C/C++:** Eigen, PCL, STL, STM32, ESP32
- Python:** PyTorch, OpenCV, Numpy
- ROS/ROS2:** Navigation2, Gazebo, Slam Toolbox, TF
- Robotics:** SLAM, VIO, EKF, ICP, Path Planning, Control
- Deep Learning:** YOLO, ViT, RNN, GAN, VAE, Diffusion, LLM
- Machine Learning:** GLM, SVM, PCA, XGBoost, HMM
- Hardware:** Circuit Design, Soldering, CAD, 3D Printing

LANGUAGES

English • Fluent (IELTS 7.5)
Mandarin • Mother tongue